

# 1 Stable outcomes in N-sided matching

## 1.1 Preference tables (N=4, n=3)

### 1.1.1 Preferences of set A

Agent  $a_1 \in A$ :

$(a_1, b_2, c_1, d_1), (a_1, b_2, c_2, d_1), (a_1, b_2, c_1, d_3), (a_1, b_1, c_2, d_2), (a_1, b_2, c_2, d_3), (a_1, b_2, c_3, d_3), (a_1, b_3, c_3, d_1),$   
 $(a_1, b_1, c_1, d_1), (a_1, b_3, c_2, d_1), (a_1, b_1, c_2, d_1), (a_1, b_3, c_1, d_1), (a_1, b_1, c_2, d_3), (a_1, b_3, c_2, d_3), (a_1, b_3, c_2, d_2),$   
 $(a_1, b_2, c_3, d_1), (a_1, b_3, c_1, d_2), (a_1, b_3, c_3, d_2), (a_1, b_2, c_2, d_2), (a_1, b_1, c_1, d_2), (a_1, b_3, c_1, d_3), (a_1, b_1, c_3, d_2),$   
 $(a_1, b_1, c_3, d_3), (a_1, b_1, c_3, d_1), (a_1, b_2, c_1, d_2), (a_1, b_2, c_3, d_2), (a_1, b_3, c_3, d_3), (a_1, b_1, c_1, d_3)$

Agent  $a_2 \in A$ :

$(a_2, b_3, c_2, d_2), (a_2, b_3, c_1, d_3), (a_2, b_2, c_2, d_3), (a_2, b_2, c_3, d_1), (a_2, b_1, c_1, d_3), (a_2, b_1, c_3, d_3), (a_2, b_3, c_2, d_3),$   
 $(a_2, b_1, c_2, d_3), (a_2, b_1, c_1, d_1), (a_2, b_3, c_2, d_1), (a_2, b_2, c_2, d_2), (a_2, b_1, c_2, d_2), (a_2, b_3, c_1, d_2), (a_2, b_1, c_3, d_2),$   
 $(a_2, b_2, c_1, d_2), (a_2, b_3, c_1, d_1), (a_2, b_3, c_3, d_2), (a_2, b_2, c_1, d_1), (a_2, b_2, c_2, d_1), (a_2, b_1, c_1, d_2), (a_2, b_2, c_3, d_2),$   
 $(a_2, b_3, c_3, d_1), (a_2, b_1, c_3, d_1), (a_2, b_1, c_2, d_1), (a_2, b_2, c_1, d_3), (a_2, b_3, c_3, d_3), (a_2, b_2, c_3, d_3)$

Agent  $a_3 \in A$ :

$(a_3, b_3, c_1, d_3), (a_3, b_3, c_2, d_2), (a_3, b_3, c_3, d_1), (a_3, b_1, c_2, d_2), (a_3, b_2, c_2, d_2), (a_3, b_1, c_2, d_3), (a_3, b_1, c_1, d_1),$   
 $(a_3, b_2, c_1, d_2), (a_3, b_2, c_2, d_1), (a_3, b_1, c_2, d_1), (a_3, b_1, c_1, d_3), (a_3, b_2, c_1, d_1), (a_3, b_2, c_3, d_3), (a_3, b_1, c_1, d_2),$   
 $(a_3, b_2, c_2, d_3), (a_3, b_1, c_3, d_3), (a_3, b_2, c_3, d_1), (a_3, b_3, c_1, d_1), (a_3, b_1, c_3, d_2), (a_3, b_3, c_1, d_2), (a_3, b_2, c_3, d_2),$   
 $(a_3, b_1, c_3, d_1), (a_3, b_3, c_3, d_2), (a_3, b_2, c_1, d_3), (a_3, b_3, c_2, d_1), (a_3, b_3, c_3, d_3), (a_3, b_3, c_2, d_3)$

### 1.1.2 Preferences of set B

Agent  $b_1 \in B$ :

$(a_2, b_1, c_2, d_2), (a_2, b_1, c_3, d_2), (a_2, b_1, c_1, d_3), (a_3, b_1, c_3, d_1), (a_2, b_1, c_2, d_3), (a_3, b_1, c_2, d_3), (a_2, b_1, c_2, d_1),$   
 $(a_1, b_1, c_2, d_2), (a_1, b_1, c_3, d_2), (a_3, b_1, c_2, d_1), (a_3, b_1, c_3, d_3), (a_2, b_1, c_1, d_2), (a_1, b_1, c_1, d_1), (a_1, b_1, c_1, d_2),$   
 $(a_2, b_1, c_3, d_1), (a_1, b_1, c_2, d_3), (a_1, b_1, c_2, d_1), (a_3, b_1, c_1, d_3), (a_3, b_1, c_1, d_1), (a_3, b_1, c_1, d_2), (a_1, b_1, c_3, d_1),$   
 $(a_3, b_1, c_2, d_2), (a_2, b_1, c_1, d_1), (a_1, b_1, c_1, d_3), (a_2, b_1, c_3, d_3), (a_3, b_1, c_3, d_2), (a_1, b_1, c_3, d_3)$

Agent  $b_2 \in B$ :

$(a_3, b_2, c_2, d_2), (a_1, b_2, c_3, d_2), (a_3, b_2, c_1, d_2), (a_1, b_2, c_2, d_1), (a_3, b_2, c_2, d_1), (a_3, b_2, c_3, d_1), (a_2, b_2, c_2, d_3),$   
 $(a_1, b_2, c_1, d_3), (a_1, b_2, c_3, d_3), (a_2, b_2, c_2, d_1), (a_1, b_2, c_2, d_3), (a_3, b_2, c_3, d_2), (a_2, b_2, c_1, d_1), (a_1, b_2, c_1, d_2),$   
 $(a_2, b_2, c_1, d_3), (a_2, b_2, c_1, d_2), (a_3, b_2, c_1, d_1), (a_3, b_2, c_2, d_3), (a_1, b_2, c_1, d_1), (a_2, b_2, c_2, d_2), (a_2, b_2, c_3, d_3),$   
 $(a_3, b_2, c_1, d_3), (a_2, b_2, c_3, d_2), (a_1, b_2, c_3, d_1), (a_2, b_2, c_3, d_1), (a_1, b_2, c_2, d_2), (a_3, b_2, c_3, d_3)$

Agent  $b_3 \in B$ :

$(a_1, b_3, c_3, d_3), (a_2, b_3, c_1, d_2), (a_2, b_3, c_3, d_2), (a_3, b_3, c_3, d_2), (a_3, b_3, c_2, d_3), (a_1, b_3, c_3, d_1), (a_2, b_3, c_2, d_2),$   
 $(a_3, b_3, c_1, d_1), (a_3, b_3, c_1, d_2), (a_3, b_3, c_3, d_3), (a_3, b_3, c_2, d_1), (a_1, b_3, c_1, d_3), (a_1, b_3, c_2, d_2), (a_2, b_3, c_2, d_3),$

$(a_3, b_3, c_2, d_2), (a_3, b_3, c_3, d_1), (a_3, b_3, c_1, d_3), (a_1, b_3, c_1, d_1), (a_1, b_3, c_2, d_3), (a_1, b_3, c_3, d_2), (a_2, b_3, c_1, d_1),$   
 $(a_2, b_3, c_2, d_1), (a_2, b_3, c_1, d_3), (a_2, b_3, c_3, d_3), (a_2, b_3, c_3, d_1), (a_1, b_3, c_1, d_2), (a_1, b_3, c_2, d_1)$

### 1.1.3 Preferences of set C

Agent  $c_1 \in C$ :

$(a_3, b_2, c_1, d_3), (a_2, b_2, c_1, d_1), (a_3, b_1, c_1, d_1), (a_3, b_3, c_1, d_1), (a_1, b_1, c_1, d_1), (a_2, b_2, c_1, d_2), (a_3, b_1, c_1, d_2),$   
 $(a_1, b_3, c_1, d_1), (a_2, b_1, c_1, d_3), (a_1, b_2, c_1, d_3), (a_2, b_3, c_1, d_3), (a_3, b_2, c_1, d_2), (a_3, b_2, c_1, d_1), (a_2, b_3, c_1, d_2),$   
 $(a_1, b_3, c_1, d_2), (a_1, b_1, c_1, d_3), (a_1, b_2, c_1, d_2), (a_3, b_3, c_1, d_2), (a_2, b_3, c_1, d_1), (a_2, b_2, c_1, d_3), (a_3, b_1, c_1, d_3),$   
 $(a_1, b_1, c_1, d_2), (a_3, b_3, c_1, d_3), (a_2, b_1, c_1, d_1), (a_1, b_3, c_1, d_3), (a_1, b_2, c_1, d_1), (a_2, b_1, c_1, d_2)$

Agent  $c_2 \in C$ :

$(a_2, b_3, c_2, d_3), (a_1, b_1, c_2, d_1), (a_1, b_1, c_2, d_2), (a_3, b_2, c_2, d_1), (a_3, b_3, c_2, d_3), (a_3, b_1, c_2, d_3), (a_2, b_1, c_2, d_2),$   
 $(a_3, b_3, c_2, d_1), (a_2, b_2, c_2, d_3), (a_3, b_1, c_2, d_2), (a_2, b_1, c_2, d_3), (a_1, b_1, c_2, d_3), (a_1, b_2, c_2, d_2), (a_1, b_2, c_2, d_3),$   
 $(a_2, b_2, c_2, d_1), (a_2, b_3, c_2, d_1), (a_3, b_1, c_2, d_1), (a_3, b_3, c_2, d_2), (a_3, b_2, c_2, d_2), (a_1, b_3, c_2, d_2), (a_2, b_3, c_2, d_2),$   
 $(a_1, b_3, c_2, d_1), (a_2, b_1, c_2, d_1), (a_3, b_2, c_2, d_3), (a_2, b_2, c_2, d_2), (a_1, b_3, c_2, d_3), (a_1, b_2, c_2, d_1)$

Agent  $c_3 \in C$ :

$(a_2, b_1, c_3, d_1), (a_3, b_3, c_3, d_3), (a_1, b_3, c_3, d_2), (a_2, b_1, c_3, d_2), (a_3, b_1, c_3, d_3), (a_3, b_1, c_3, d_2), (a_3, b_2, c_3, d_1),$   
 $(a_3, b_2, c_3, d_3), (a_2, b_2, c_3, d_2), (a_1, b_2, c_3, d_2), (a_1, b_3, c_3, d_1), (a_3, b_3, c_3, d_2), (a_1, b_1, c_3, d_2), (a_2, b_1, c_3, d_3),$   
 $(a_2, b_3, c_3, d_3), (a_3, b_1, c_3, d_1), (a_1, b_3, c_3, d_3), (a_1, b_1, c_3, d_3), (a_2, b_3, c_3, d_1), (a_3, b_3, c_3, d_1), (a_1, b_2, c_3, d_1),$   
 $(a_3, b_2, c_3, d_2), (a_1, b_1, c_3, d_1), (a_2, b_3, c_3, d_2), (a_2, b_2, c_3, d_1), (a_1, b_2, c_3, d_3), (a_2, b_2, c_3, d_3)$

### 1.1.4 Preferences of set D

Agent  $d_1 \in D$ :

$(a_3, b_2, c_2, d_1), (a_3, b_3, c_1, d_1), (a_1, b_3, c_3, d_1), (a_1, b_1, c_3, d_1), (a_2, b_3, c_2, d_1), (a_2, b_2, c_2, d_1), (a_1, b_3, c_1, d_1),$   
 $(a_2, b_1, c_3, d_1), (a_2, b_3, c_3, d_1), (a_2, b_1, c_2, d_1), (a_3, b_1, c_2, d_1), (a_1, b_1, c_2, d_1), (a_3, b_1, c_3, d_1), (a_3, b_2, c_1, d_1),$   
 $(a_1, b_1, c_1, d_1), (a_3, b_1, c_1, d_1), (a_1, b_3, c_2, d_1), (a_3, b_3, c_3, d_1), (a_3, b_2, c_3, d_1), (a_2, b_2, c_3, d_1), (a_1, b_2, c_3, d_1),$   
 $(a_3, b_3, c_2, d_1), (a_1, b_2, c_1, d_1), (a_1, b_2, c_2, d_1), (a_2, b_3, c_1, d_1), (a_2, b_2, c_1, d_1), (a_2, b_1, c_1, d_1)$

Agent  $d_2 \in D$ :

$(a_3, b_2, c_3, d_2), (a_3, b_2, c_2, d_2), (a_3, b_1, c_2, d_2), (a_2, b_3, c_3, d_2), (a_1, b_2, c_3, d_2), (a_3, b_3, c_3, d_2), (a_2, b_2, c_1, d_2),$   
 $(a_2, b_1, c_1, d_2), (a_2, b_3, c_1, d_2), (a_3, b_2, c_1, d_2), (a_2, b_1, c_2, d_2), (a_3, b_3, c_1, d_2), (a_1, b_1, c_2, d_2), (a_1, b_1, c_1, d_2),$   
 $(a_1, b_2, c_2, d_2), (a_2, b_2, c_3, d_2), (a_2, b_2, c_2, d_2), (a_3, b_1, c_3, d_2), (a_1, b_1, c_3, d_2), (a_3, b_1, c_1, d_2), (a_1, b_2, c_1, d_2),$   
 $(a_3, b_3, c_2, d_2), (a_1, b_3, c_2, d_2), (a_2, b_1, c_3, d_2), (a_2, b_3, c_2, d_2), (a_1, b_3, c_3, d_2), (a_1, b_3, c_1, d_2)$

Agent  $d_3 \in D$ :

$(a_3, b_2, c_2, d_3), (a_2, b_3, c_2, d_3), (a_2, b_3, c_3, d_3), (a_1, b_3, c_3, d_3), (a_1, b_3, c_1, d_3), (a_1, b_1, c_1, d_3), (a_3, b_1, c_3, d_3),$   
 $(a_2, b_2, c_2, d_3), (a_3, b_1, c_1, d_3), (a_2, b_1, c_1, d_3), (a_1, b_2, c_2, d_3), (a_3, b_3, c_2, d_3), (a_1, b_2, c_3, d_3), (a_3, b_3, c_3, d_3),$

$(a_1, b_3, c_2, d_3), (a_2, b_2, c_1, d_3), (a_3, b_2, c_1, d_3), (a_2, b_1, c_2, d_3), (a_3, b_2, c_3, d_3), (a_1, b_2, c_1, d_3), (a_1, b_1, c_2, d_3),$   
 $(a_3, b_3, c_1, d_3), (a_3, b_1, c_2, d_3), (a_2, b_2, c_3, d_3), (a_2, b_3, c_1, d_3), (a_2, b_1, c_3, d_3), (a_1, b_1, c_3, d_3)$

$$\begin{array}{l}
\mu_1: (a_1, b_1, c_2, d_2), (a_2, b_2, c_1, d_1), (a_3, b_3, c_3, d_3) \\
\mu_2: (a_1, b_1, c_2, d_2), (a_2, b_3, c_1, d_3), (a_3, b_2, c_3, d_1) \\
\mu_3: (a_1, b_2, c_3, d_2), (a_2, b_1, c_2, d_3), (a_3, b_3, c_1, d_1) \\
\mu_4: (a_1, b_2, c_3, d_2), (a_2, b_3, c_2, d_3), (a_3, b_1, c_1, d_1) \\
\mu_5: (a_1, b_3, c_1, d_3), (a_2, b_1, c_3, d_1), (a_3, b_2, c_2, d_2) \\
\mu_6: (a_1, b_3, c_3, d_1), (a_2, b_1, c_1, d_3), (a_3, b_2, c_2, d_2) \\
\mu_7: (a_1, b_3, c_3, d_1), (a_2, b_2, c_1, d_2), (a_3, b_1, c_2, d_3) \\
\mu_8: (a_1, b_3, c_3, d_2), (a_2, b_1, c_1, d_3), (a_3, b_2, c_2, d_1)
\end{array}$$

Table 1: Stable matches for N=4, n=3

## 1.2 Stable matches

For our example in N-sided matching ( $N = 4, n = 3$ ), our algorithm produces exactly eight stable matches (see table 1).